

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (CSE) (Credit System) Degree Examinations

April 2023

21CS304 - DIGITAL SYSTEMS DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Provide examples and diagrams necessary.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Construct EX-NOR gate using NAND & NOR gates.	10	L*3	2	1,2,3
b)	Solve the given min term expression using Quine McClusky method. $F(A,B,C,D) = \sum m(0,1,2,4,6,8,9,11,13,15)$	10	L3	1	1,2,3
2. a)	Construct AND, OR and NOT gate using NOR gates.	6	L3	2	1,2,3
b)	Solve the given maxterm expression using Quine McClusky method. $F(a,b,c,d) = \prod m(0,1,2,5,6,7,8,9,10,14)$.	10	L3	1	1,2,3
c)	Simplify following expression using K-map and realize using logic gates. $f(a,b,c,d) = \sum m(0,1,2,5,6,7,8,9,10,14)$.	4	L3	2	1,2,3
3. a)	Simplify the following Boolean function. i) $F = ABC + BCD + ABC + ABCD$ ii) $F(A,B,C,D) = \sum m(1,3,4,5,6,7,12,14)$	8	L3	1	1,2,3
b)	Convert the following given binary to decimal. i) 1001 ii) 1010 iii) 1101 iv) 0101	4	L3	2	1,2,3
c)	Simplify following expression using K-map and realize using logic gates. $f(a,b,c,d) = \sum m(0,1,5,6,8,9,11,13) + \sum d(7,10,12)$	8	L3	1	1,2,3
Unit – II					
4. a)	Design and explain full adder in detail.	10	L3	3	1,2,3
b)	Construct and explain the 3:8 Decoder with neat diagram.	10	L3	3	1,2,3
5. a)	Construct 16:1 multiplexer using 4:1 mux.	8	L3	3	1,2,3
b)	Realize 2:1 mux using verilog code HDL implementation.	6	L3	4	1,2,3
c)	Illustrate the working of edge triggered JK flipflop.	6	L3	4	1,2,3
6. a)	Explain the working of encoder in detail.	10	L3	3	1,2,3
b)	Explain the working of T-flipflop and generate the characteristic equation.	10	L3	3	1,2,3
Unit – III					
7. a)	Explain the working of Serial In Parallel Out shift register.	6	L5	5	1,2,3
b)	Realize JK FF using Verilog HDL implementation.	6	L5	5	1,2,3
c)	Design Mod-5 synchronous up counter with neat diagram.	8	L5	5	1,2,3
8. a)	Compare and contrast between synchronous and asynchronous counter.	10	L5	5	1,2,3
b)	Implement D-flipflop using Verilog implementation.	10	L5	5	1,2,3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
